

22SLC3

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Problem

In each square of a garden shaped like a 2022×2022 board, there is initially a tree of height 0. A gardener and a lumberjack alternate turns playing the following game, with the gardener taking the first turn:

- The gardener chooses a square in the garden. Each tree on that square and all the surrounding squares (of which there are at most eight) then becomes one unit taller.
- The lumberjack then chooses four different squares on the board. Each tree of positive height on those squares then becomes one unit shorter.

We say that a tree is *majestic* if its height is at least 10^6 . Determine the largest K such that the gardener can ensure there are eventually K majestic trees on the board, no matter how the lumberjack plays.

Solution. We claim that $K = \frac{5}{9} \cdot 2022^2$.

The strategy for the gardener goes as follows:

Divide the 2022×2022 into multiple 3×3 squares. Since $3 \mid 2022$, we can do that. Now for each 3×3 square, the gardener plants a tree on all of the squares. He plants a total of 9 trees. The lumberjack can reduce at most 4 trees from the 9 planted trees. As a result, there is a net "gain" of 5 units of height everytime the lumberjack adds a height to all of the 9 trees. Since the lumberjack can do this arbitrary many times, for all $N \in \mathbb{N}$, he can make at least 5 of the trees in that 3×3 area to have a height $> N$. Say he does this in order for every 3×3 grid and he makes the height of each grid at least $n_1, n_2, \dots, n_{674^2}$ where n_i is the i th grid where he does this move arbitrarily. Note that in this strategy, the lumberjack does all his moves for each grid "at once", i.e., he won't return to that grid after he moves on to another grid. We will decide the values of $n_1, n_2, \dots, n_{674^2}$ later. Replace 10^6 with M . Observe that if we do M^2 moves for any 3×3 squares (ignoring the others), it will be enough for at least 5 trees to have a height of M .

Consider the sequence $a_k = M^{2^{k-1}}$ with $a_1 = M^2$. Then consider $n_i = a_{674^2-i+1}$. Since at any given point, the total "extra" height in the 5 trees of n_k is greater than $n_{k+1} + n_{k+2} + \dots + n_{674^2}$, the gardener can ensure at least $\frac{5}{9} \cdot 2022^2$.

The strategy for the lumberjack goes as follows:

For any given 3×3 square, color the 4 corners red. Now, for any 3×3 grid that the gardener chooses, observe that you will have at most 4 red squares where he grows the trees. The lumberjack will remove the height from the trees in the red squares for sure. Using this strategy, he can ensure that there is never any net "gain" in height in any of the red squares. Since we have a total of $\frac{4}{9} \cdot 2022^2$ red squares, he can ensure that all those squares have no trees at all. So the gardener can have at most $\frac{5}{9} \cdot 2022^2$ trees.

Since we know that the lumberjack can prevent more than $\frac{5}{9} \cdot 2022^2$ trees and the gardener can ensure at least $\frac{5}{9} \cdot 2022^2$ trees, we can conclude that $K = \frac{5}{9} \cdot 2022^2$.